

Regularization and Stability

Statistical Methods for Machine Learning Project

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1 Introduction

This project investigates the relationship between regularization strength (λ), algorithmic stability, and generalization performance (test error).

Classical learning theory explains why models generalize using tools like uniform convergence and complexity measures, and motivates regularization as a way to prevent overfitting. But this view doesn't fully explain why regularized algorithms generalize well.

Bousquet and Elisseeff (2002) propose stability as the mechanism behind that; an algorithm is stable if small changes to the training set don't significantly change its predictions. Their claim is that stability, not hypothesis class complexity, is what actually causes generalization.

To test this, Ridge regression was implemented and a leave-one-out stability metric was defined; remove each training point, retrain, and measure how much the test predictions change. The hypothesis is that increasing λ increases stability, and that higher stability correlates with better generalization.

This was evaluated on two datasets: a synthetic one with known ground truth weights, and a real world one (UCI Concrete Compressive Strength) to check if the pattern holds both in a controlled setting and in practice. Three extensions were also added, which are comparing Ridge to unregularized least squares, testing how training set size affects stability, and comparing L1 (Lasso) and L2 (Ridge) regularization.

2 Datasets

2.1 Synthetic Dataset

A synthetic regression dataset was generated to evaluate Ridge regression under controlled, known conditions. Inputs were sampled from a standard Gaussian distribution: $X \in \mathbb{R}^{n \times d}$ with $n = 500$ and $d = 8$. Labels were generated as a linear function of the inputs with additive Gaussian noise:

$$y = Xw^* + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (1)$$

A **sparse ground-truth weight vector** was chosen:

$$w^* = [1.5, -2.0, 0.8, 0.0, 0.0, 0.0, 1.2, 0.0] \quad (2)$$

with 4 informative features and 4 irrelevant features. This choice was made to support the L1 vs L2 extension (Section 5.3); it allows us to evaluate whether Lasso correctly identifies the irrelevant features, which would not be observable if all features were informative.

The noise level was set to $\sigma = 2.0$. First experiments were done with $\sigma = 0.5$, but it was found that the resulting dataset was too close to a noiseless linear relationship. Ridge regression fit the data almost perfectly regardless of λ , and the effect of regularization on both test error and stability was too small to observe clearly. Increasing the noise to $\sigma = 2.0$ produced a more realistic signal-to-noise ratio in which the effect of regularization becomes clearly observable, while still preserving the dataset’s main purpose. Data was split into 80% training (400 points) and 20% test (100 points).

2.2 Concrete Compressive Strength Dataset

For the real-world dataset, the UCI Concrete Compressive Strength dataset (which will be referred as concrete dataset) was used, which contains 1030 instances with 8 quantitative input features (cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age) and one continuous target variable (compressive strength in MPa). This dataset was chosen because it is a clean regression problem of a moderate size suitable for repeated retraining in the stability experiments.

Preprocessing: The dataset’s features span very different scales (e.g., cement is measured in the hundreds, superplasticizer in single digits). Since the Ridge penalty $\lambda\|w\|^2$ applies uniformly across all weights, features on larger scales would otherwise dominate the penalty term. So all features (zero mean, unit variance) and the target variable were standardized using the statistics of the training set, to avoid data leakage.

Shuffling: It was observed that the target variable was not uniformly distributed across the dataset’s original ordering. A train/test split without shuffling produced an anomalous result (train error higher than test error), indicating that the train and test sets were picked from different underlying distributions. So a random permutation was applied before splitting the data. Data was split into 80% training (824 points) and 20% test (206 points), like the synthetic data.

3 Methodology

3.1 Ridge Regression

Ridge regression is the solution to the regularized least squares problem:

$$\min_w \sum_{i=1}^n (y_i - w^\top x_i)^2 + \lambda \|w\|_2^2 \quad (3)$$

This has a closed-form solution:

$$w = (X^\top X + \lambda I)^{-1} X^\top y \quad (4)$$

This was implemented directly by solving the linear system $(X^\top X + \lambda I)w = X^\top y$ rather than explicitly computing the matrix inverse. Unregularized least squares was obtained as the special case $\lambda = 0$.

3.2 Stability Metric

Following Bousquet and Elisseeff, stability was defined with a leave-one-out procedure. For a fixed training set, test set, and regularization strength λ , we train the model on the full training set to obtain test predictions $\hat{y}$. Then, for each training point i , it is removed, the model is retrained on

the remaining points to obtain new test predictions $\hat{y}^{(-i)}$, and the resulting average absolute change in predictions on the test set was computed. The stability score $S(\lambda)$ is defined as the average of this change across all training points. A lower value indicates a more stable algorithm. Removing any single training point has, on average, less effect on the model’s predictions on unseen data.

Stability was measured with the change in predictions rather than the change in loss, since this directly and interpretably captures how much the model’s output changes, without mixing it up with the size of the loss function itself. This change was evaluated on the test set rather than the training set, since the goal is to measure the effect on generalization behavior, not fit to the training data.

3.3 Lasso with Coordinate Descent

For the L1 vs L2 extension, Lasso regression was implemented, which replaces the L2 penalty with an L1 penalty: $\|w\|_1$ instead of $\|w\|_2^2$. Unlike Ridge, this objective has no closed-form solution, since the L1 penalty is non-differentiable at $w_j = 0$. So coordinate descent was used, a standard approach for L1 regularized problems. At each iteration, every coordinate w_j is updated in turn while holding all other coordinates fixed, using the soft-thresholding operator, which shrinks each coordinate toward zero and sets it exactly to zero once its contribution falls below the regularization threshold λ . This coordinate-wise update is repeated until the change in weights between iterations falls below a small tolerance, or a maximum number of iterations is reached.

4 Results

4.1 Stability and Error vs. Regularization Strength — Synthetic Dataset

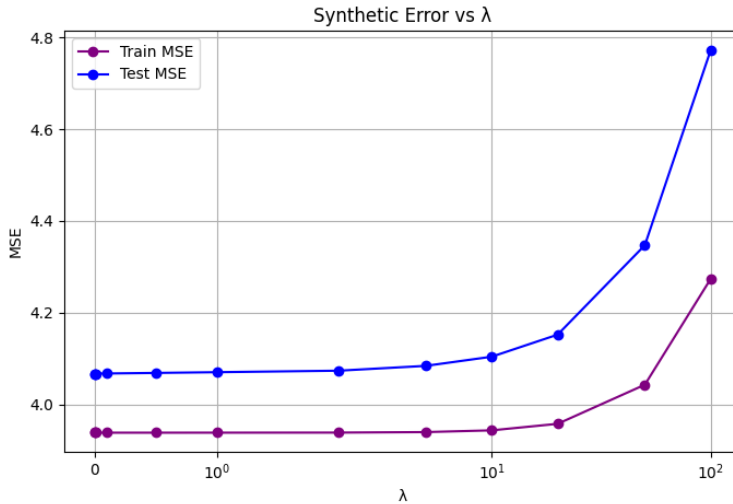


Figure 1: Synthetic dataset: Train/Test MSE vs. λ .

For the synthetic dataset, training error increases with λ as expected (Figure 1). Stronger regularization constrains the model away from the unregularized least-squares fit. Test error remains nearly flat for small λ ($\lambda \leq 10$) and then increases for larger λ , producing the expected U-shape, even though in our experiments the increase past the minimum is much steeper than the decrease

leading into it. This is because the synthetic data is close enough to the model’s assumptions that under-regularization is barely penalized, while over-regularization is.

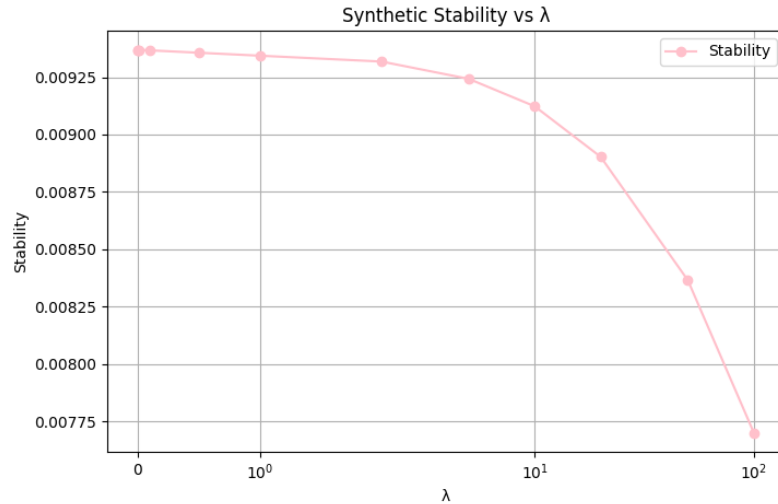


Figure 2: Synthetic dataset: Stability vs. λ .

The stability curve is decreasing across the full tested range (Figure 2). Stability score drops from approximately 0.0094 at $\lambda = 0$ to approximately 0.0077 at $\lambda = 100$. This supports the core hypothesis cleanly on the synthetic dataset: increasing regularization increases stability.

4.2 Stability and Error vs. Regularization Strength — Concrete Dataset

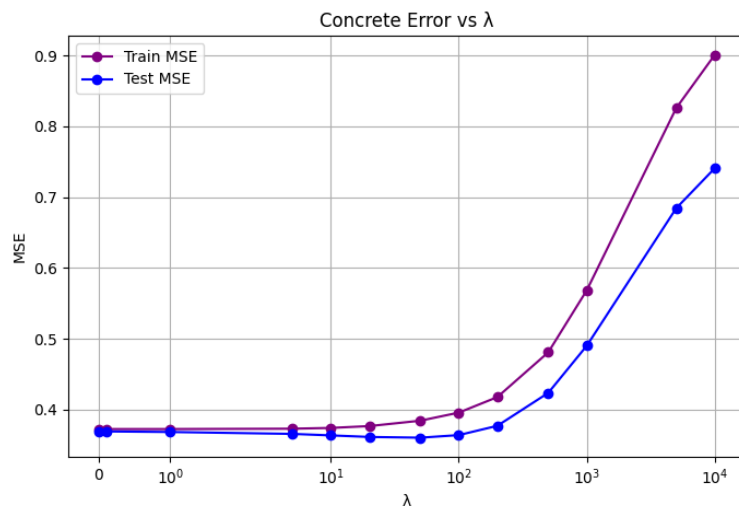


Figure 3: Concrete dataset: Train/Test MSE vs. λ .

Figure 3 shows that on the Concrete dataset, the same pattern continues but more clearly. Training error increases monotonically with λ . Test error shows a clear U-shape, decreasing from $\lambda = 0$ (test

MSE ≈ 0.369) to a minimum around $\lambda = 50$ (test MSE ≈ 0.360), then increasing again for larger λ (test MSE ≈ 0.741 at $\lambda = 10000$).

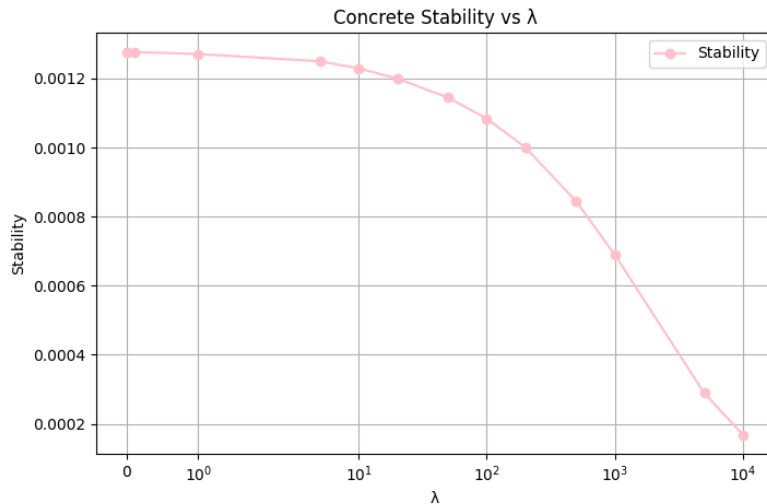


Figure 4: Concrete dataset: Stability vs. λ .

The stability score decreases monotonically across the entire range tested (Figure 4; $\lambda = 0$ to $\lambda = 10000$), from approximately 0.00128 to 0.00017. Unlike the synthetic case, no instability re-emerges at large λ within the range we tested. This is the cleanest confirmation of the core hypothesis in our experiments: regularization strength and stability are inversely related across the full range studied.

4.3 Test Error vs. Stability

To directly test the second part of the hypothesis that more stable models generalize better, test error was plotted against the stability score itself, using λ as an implicit parameter along the curve.

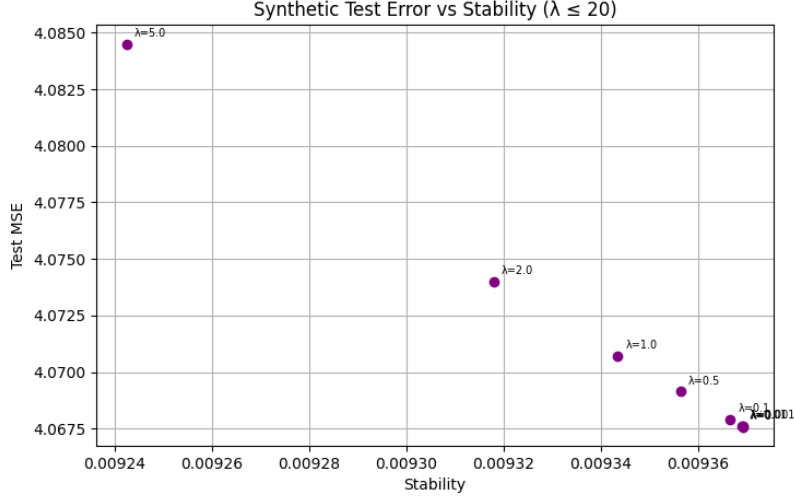


Figure 5: Concrete dataset: Test error vs. stability score, annotated by λ .

Figure 5 shows that on the concrete dataset, this relationship is strong and monotonic for the range $\lambda \leq 200$: as stability improves (lower stability score), test error decreases. Beyond this range ($\lambda > 200$), the relationship breaks down, stability continues to improve (lower score) but test error worsens. This is because at very large λ , the model is driven toward underfitting. Its predictions become highly consistent regardless of which training point is removed, simply because the model has learned very little from the data in the first place. This illustrates an important limitation of the stability generalization link. It holds only when the model still fits the data reasonably well and is not severely underfitting. Stability is associated with good generalization, but it is not sufficient on its own.

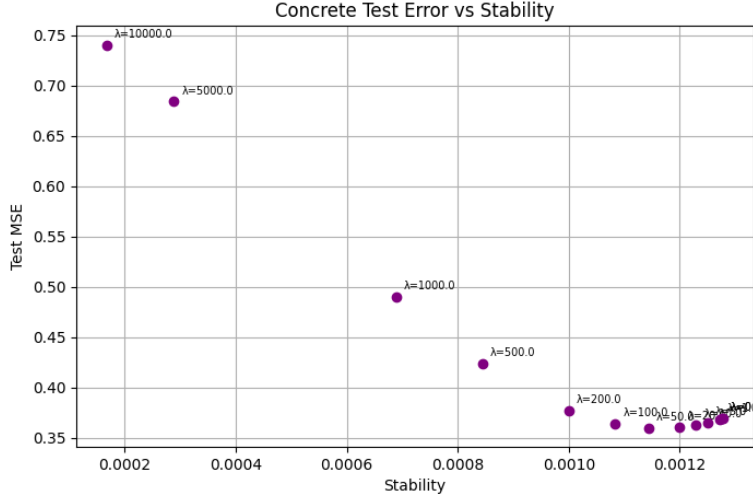


Figure 6: Synthetic dataset: Test error vs. stability score, annotated by λ ($\lambda \leq 20$).

Figure 6 shows that on the synthetic dataset, the plot was restricted to $\lambda \leq 20$ for the same reason discussed in Section 4.1. Within this range, the relationship between stability and test error

is also positive, though the dataset’s narrow range of stability values (0.0092–0.0094) makes the trend visually less dramatic than on concrete dataset.

4.4 Ridge vs. Unregularized Least Squares

Unregularized least squares corresponds to the $\lambda = 0$ point on all curves above. On both datasets, $\lambda = 0$ produces the highest stability score and is not the point of minimum test error, confirming that some amount of regularization improves both generalization and stability relative to the unregularized baseline, though the improvement is modest for the synthetic dataset and more visible for the concrete dataset.

4.5 Effect of Dataset Size on Stability

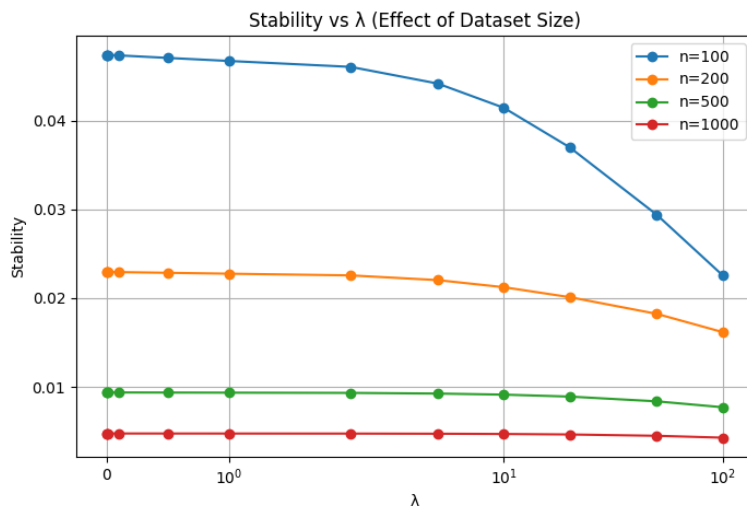


Figure 7: Synthetic dataset: Stability vs. λ for varying training set size.

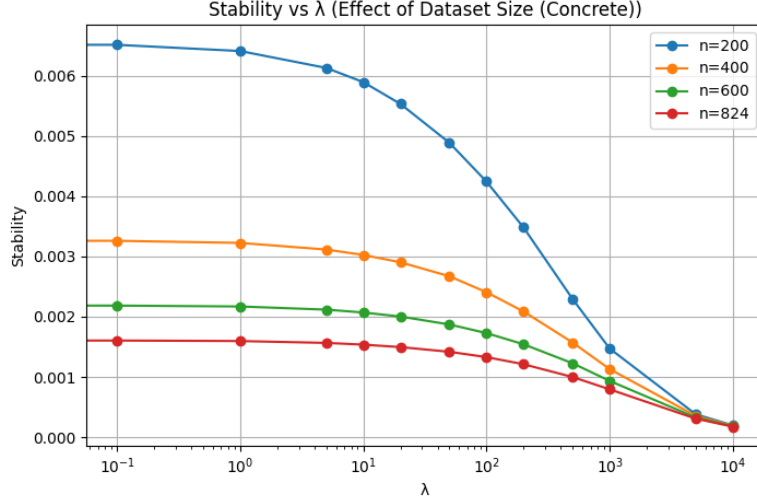


Figure 8: Concrete dataset: Stability vs. λ for varying training set size.

The stability experiment was repeated for varying training set sizes, keeping other settings fixed. On both datasets (Figures 7 and 8), stability score decreases as n increases, for every value of λ tested. On the synthetic dataset, stability at $\lambda = 0$ drops from approximately 0.047 ($n = 100$) to approximately 0.005 ($n = 1000$) — nearly a 10x improvement. On the concrete dataset, the same pattern continues.

This result is consistent with the theoretical intuition: removing a single point from a larger dataset represents a smaller relative change to the training distribution, and so has a proportionally smaller effect on the learned model. It was also observed that on the concrete dataset, the curves for different n converge as λ becomes very large. At extreme regularization, dataset size becomes less relevant because the model is dominated by the penalty term rather than the data.

4.6 L1 vs. L2 Regularization

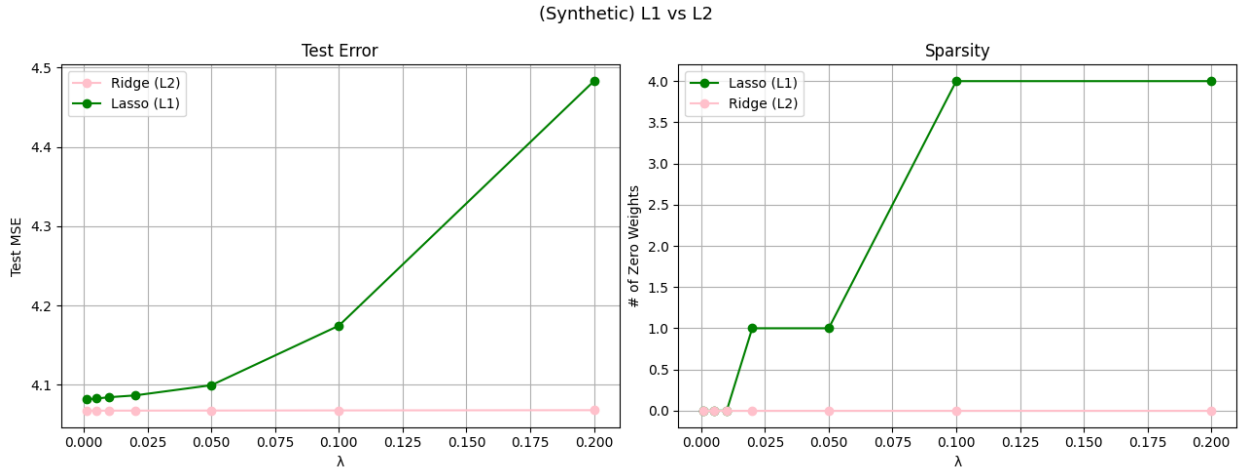


Figure 9: Synthetic dataset: Test error and sparsity, Ridge (L2) vs. Lasso (L1).

Figure 9 shows that on the synthetic dataset, where the ground truth contains 4 zero weight features, Lasso correctly drives weights to zero as λ increases, reaching all 4 correct zeros by $\lambda = 0.1$. However, in this λ range Lasso’s test error is already higher than Ridge’s, which remains flat for the entire tested range. This hints a trade off, by the time Lasso’s regularization is strong enough to correctly zero out the irrelevant features, it is also strong enough to shrink the informative weights enough to hurt predictive performance on this dataset.

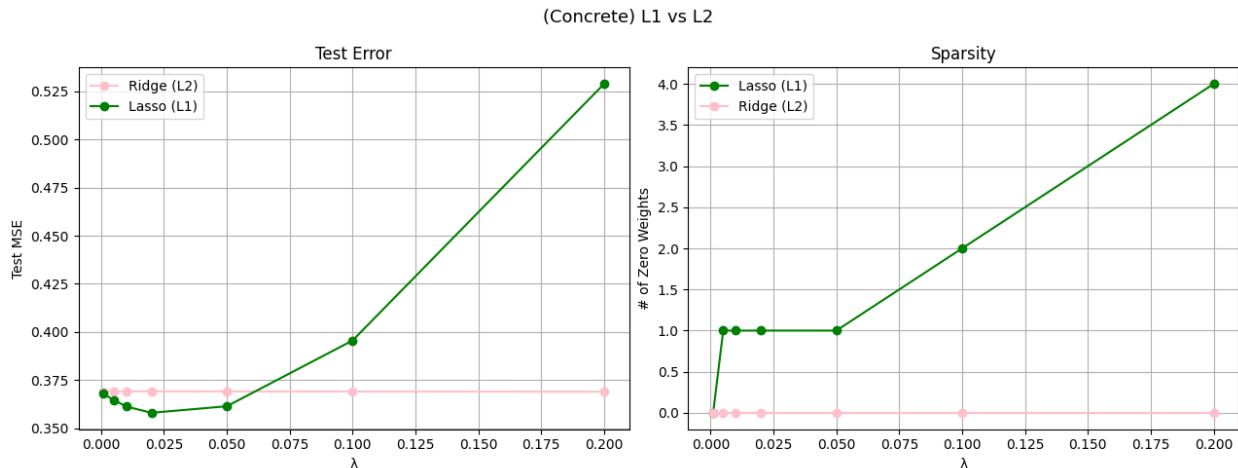


Figure 10: Concrete dataset: Test error and sparsity, Ridge (L2) vs. Lasso (L1).

Figure 10 shows that on the Concrete dataset, the result is better than Lasso. At $\lambda = 0.02$, Lasso achieves a lower test error (0.358) than Ridge (0.369) while setting one feature’s weight to zero. This suggests that the Concrete dataset contains at least one feature that contributes little predictive value, which Lasso is able to identify and show, whereas Ridge never sets weights to exactly zero.

5 Discussion

5.1 Summary of Findings

The experiments provided empirical support for the central hypothesis of Bousquet and Elisseeff: increasing regularization strength increases algorithmic stability, and within a range of λ increased stability correlates with improved generalization. This was visible on the concrete dataset and, more weakly but still visibly on the synthetic dataset.

However, the results also revealed an important limitation of this relationship. The positive correlation between stability and test error is true only when the model is still reasonably well fitted. Once λ becomes large enough to cause underfitting, stability continues to improve monotonically, the model becomes increasingly insensitive to individual training points, since it is relying less and less on the data, while test error worsens. This shows stability is necessary but not sufficient for good generalization. A model can be perfectly stable simply by ignoring the data and always predicting the same value but such a model is not useful, since it fails to capture the actual relationship between inputs and outputs.

5.2 Comparing the Two Datasets

The concrete dataset produced a substantially cleaner confirmation of the stability generalization relationship than the synthetic dataset. This is caused by two factors. First, the Concrete dataset is real world data with a more complex relationship between features and target than the synthetic dataset, leaving more room for regularization to meaningfully trade off bias and variance. Second, the synthetic dataset was generated to closely satisfy Ridge regression’s modeling assumptions (linear relationship, Gaussian noise, all relevant features present at comparable scale), which limits how much regularization can help. A model that is already close to the ”true” data generating process has less to gain from additional regularization. This shows that the predictions of stability theory are most clearly visible when the model does not perfectly match the data.

5.3 Limitations

Several limitations can be noted:

Stability metric choice: Average change in test predictions was used as the stability metric; using average change in loss instead might have given different results, particularly in the underfitting regime discussed in Section 5.1.

Computational cost: The stability metric we used requires retraining the model once per training point, so it might not scale well to much larger datasets.

Synthetic dataset realism: The synthetic dataset remains a best-case scenario for Ridge regression by construction. This is viewed as appropriate for verifying the theory under controlled conditions, but it limits how much the synthetic results alone can say about real-world performance.

6 Conclusion

This project empirically investigated the relationship between regularization strength, algorithmic stability, and generalization performance in linear regression, following the theoretical framework of Bousquet and Elisseeff. Ridge regression and a leave-one-out stability metric was implemented, and they were evaluated on both a synthetic dataset with known ground truth structure and a real-world Concrete Compressive Strength dataset.

The results confirm that increasing regularization strength increases stability, within the range where the model is not severely underfit, and increased stability is associated with improved generalization. This relationship was most clearly visible on the real-world dataset. Also an important limit to this theory was seen; even after the model starts underfitting and its test error gets worse, its stability score keeps improving. This shows that stability on its own does not guarantee good generalization. A model can become more consistent without becoming more accurate.

Comparing Ridge to unregularized least squares confirmed that the unregularized model was both the least stable and not the best generalizing configuration on either dataset. Training set size was shown to have a strong effect on stability, with larger datasets producing more stable models at every regularization level tested. Finally, comparing L1 and L2 regularization showed that Lasso’s advantage in correctly identifying irrelevant features is dependant on the dataset. It improved performance on the concrete dataset, but not on the synthetic dataset, where the regularization needed to correctly zero out irrelevant features was already strong enough to hurt the informative weights.

Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

References

Bousquet, O., & Elisseeff, A. (2002). Stability and Generalization. *Journal of Machine Learning Research*, 2, 499–526.

Yeh, I. (1998). Concrete Compressive Strength [Dataset]. UCI Machine Learning Repository. <https://archive-beta.ics.uci.edu/dataset/165/concrete+compressive+strength>